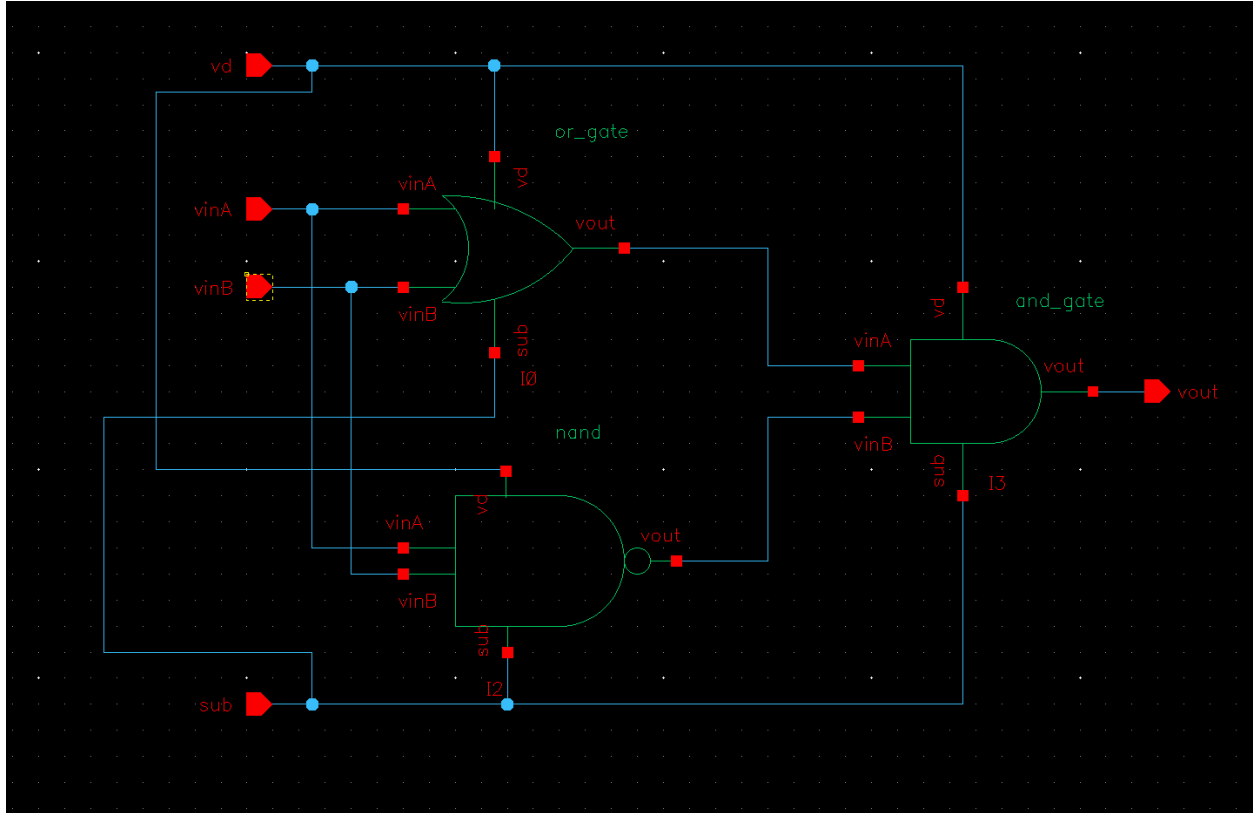
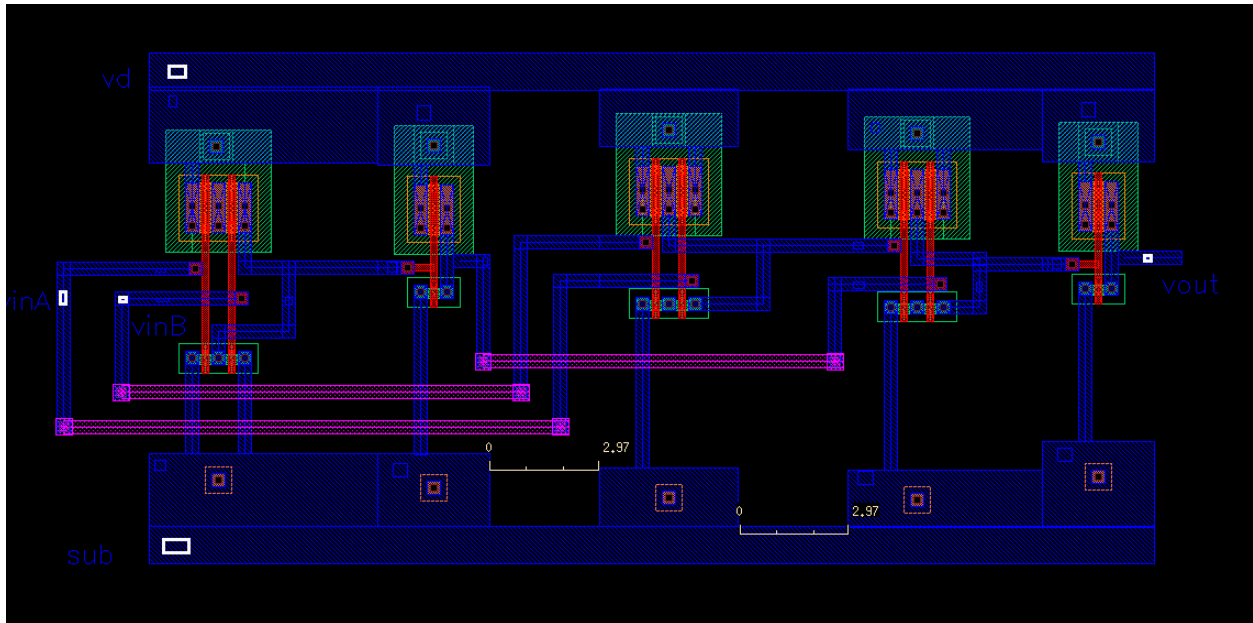


Project 2 Part3 - Block Layout
Gerald Stanford

Xor Gate:



Schematic



layout

```
DRC started at Wed Dec 6 21:00:49 2023

Validating hierarchy instantiation for:
library: Stanford
cell:    xor_gate
view:    layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaDRC.rul.
Inclusion limit is set to 1000.
Running layout DRC analysis
Flat mode
Full checking.
DRC started.....Wed Dec 6 21:00:49 2023
completed ....Wed Dec 6 21:00:49 2023
CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
***** Summary of rule violations for cell "xor_gate layout" *****
Total errors found: 0
```

Drc

```

Extraction started at Wed Dec 6 21:03:51 2023

Validating hierarchy instantiation for:
library: Stanford
cell:    xor_gate
view:    layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaEXT.rul.
Inclusion limit is set to 1000.
Running layout Extraction analysis
flat mode
Full checking.
For layer nChannelTran :
    8 shapes encountered.
    8 nmos4 ivpcell NCSU_Analog_Parts devices well formed.
For layer pChannelTran :
    8 shapes encountered.
    8 pmos4 ivpcell NCSU_Analog_Parts devices well formed.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
Extraction started.....Wed Dec 6 21:03:51 2023
    completed ...Wed Dec 6 21:03:52 2023
    CPU TIME = 00:00:00 TOTAL TIME = 00:00:01
***** Summary of rule violations for cell "xor_gate layout" *****
    Total errors found: 0

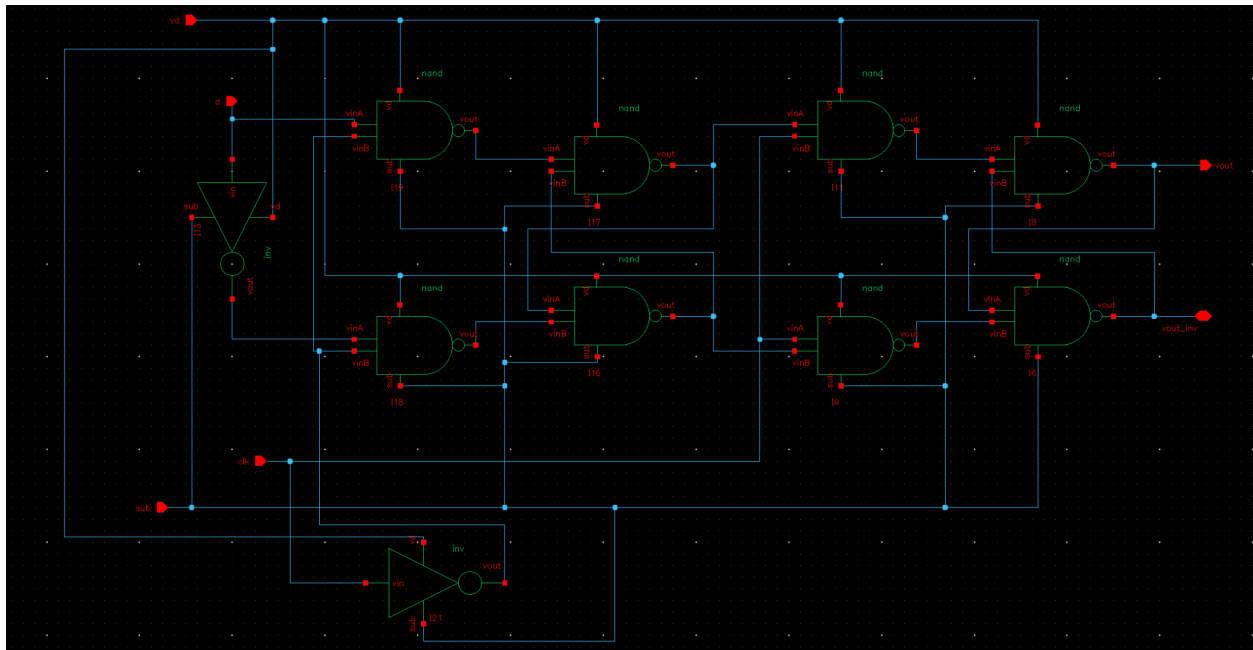
saving rep Stanford/xor_gate/extracted
Getting layout proptert bagGetting layout proptert bag
*Error* eval: unbound variable - ivLVSContentsDifferForm
*WARNING* ddGetObj: cell name 'xor_gate ' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate ' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate ' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate ' is not a legal cell name.
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/eceguest/digital_course/LVS

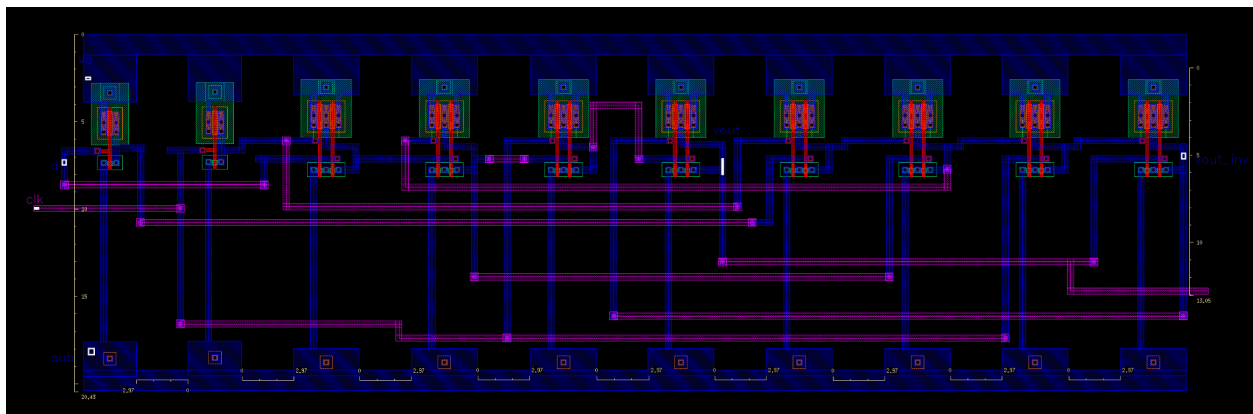
```

LVS

D-FlipFlop:



Schematic



Layout

DRC started at Wed Dec 6 21:10:22 2023

Validating hierarchy instantiation for:

library: Stanford

cell: D_flipflop

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaDRC.rul.

Inclusion limit is set to 1000.

Running layout DRC analysis

Flat mode

Full checking.

DRC started.....Wed Dec 6 21:10:22 2023

completedWed Dec 6 21:10:22 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:00

***** Summary of rule violations for cell "D_flipflop layout" *****

Total errors found: 0

DRC

Extraction started at Wed Dec 6 21:11:06 2023

Validating hierarchy instantiation for:

library: Stanford

cell: D_flipflop

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaEXT.rul.

Inclusion limit is set to 1000.

Running layout Extraction analysis

flat mode

Full checking.

For layer nChannelTran :

18 shapes encountered.

18 nmos4 ivpcell NCSU_Analog_Parts devices well formed.

For layer pChannelTran :

18 shapes encountered.

18 pmos4 ivpcell NCSU_Analog_Parts devices well formed.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

Extraction started.....Wed Dec 6 21:11:06 2023

completedWed Dec 6 21:11:07 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "D_flipflop layout" *****

Total errors found: 0

saving rep Stanford/D_flipflop/extracted

Getting layout propert bagGetting layout propert bag

***Error* eval: unbound variable - ivLVSContentsDifferForm**

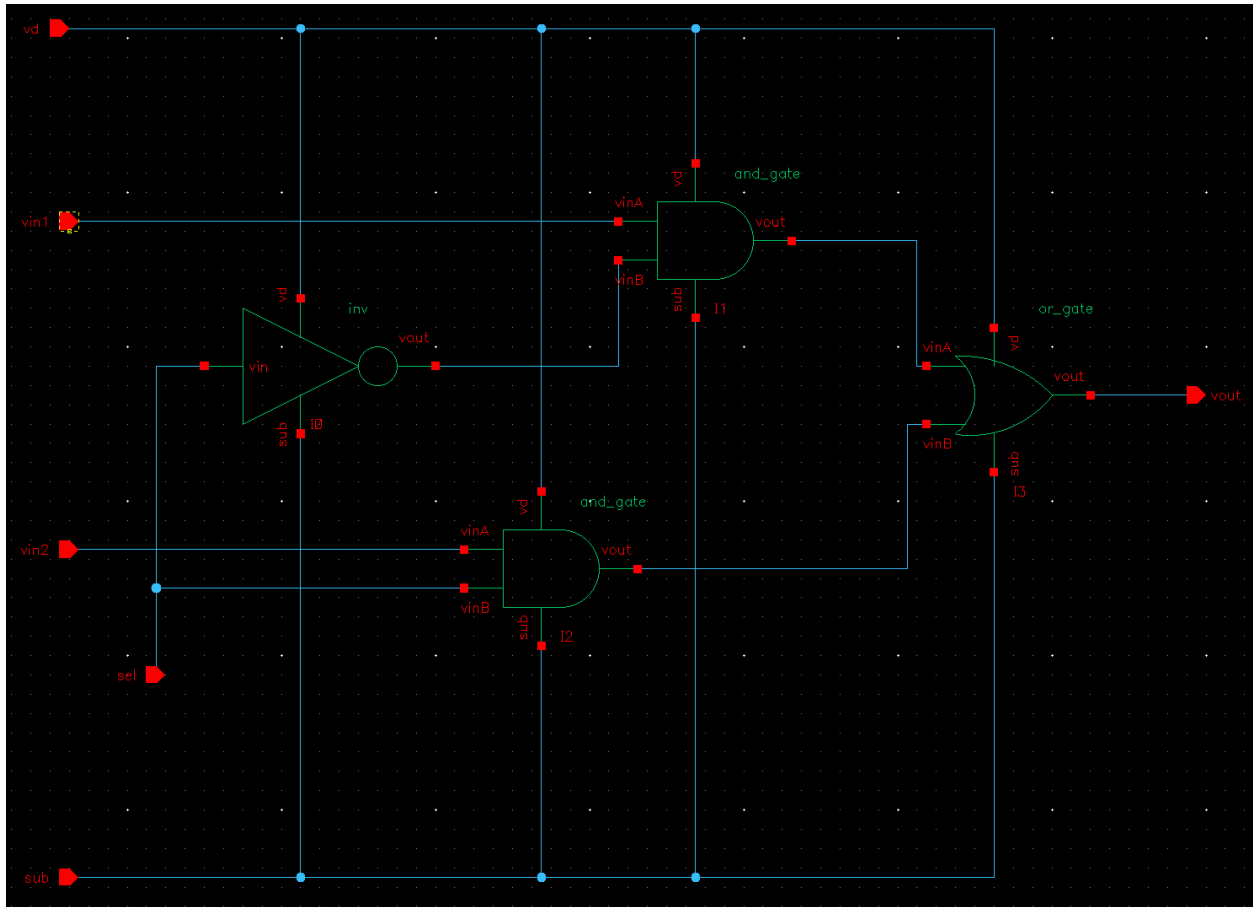
LVS job is now started...

The LVS job has completed. The net-lists match.

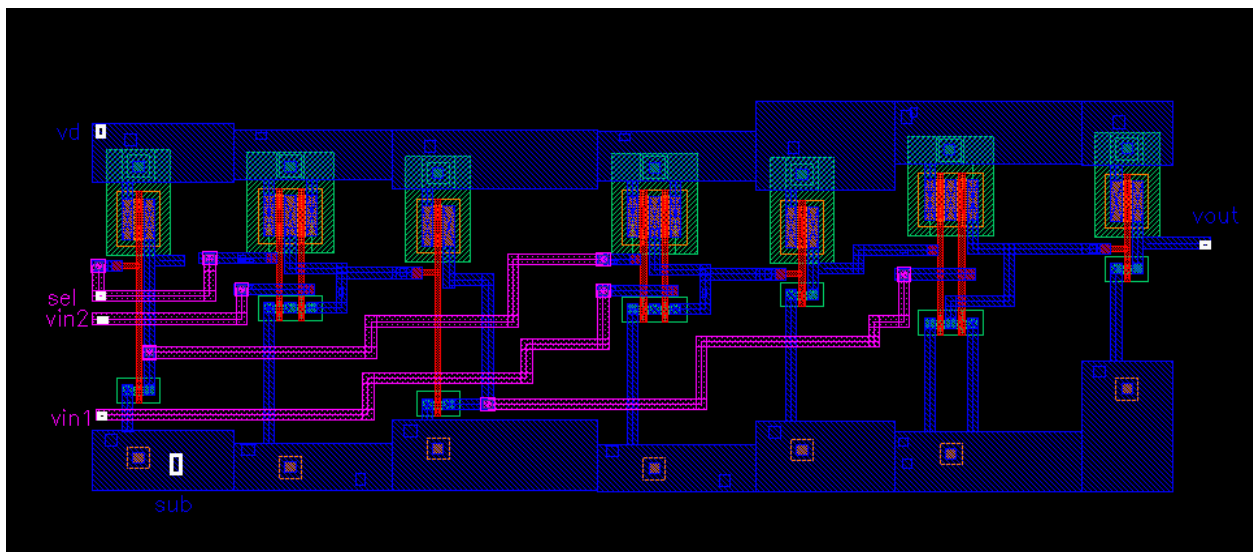
Run Directory: /home/eceguest/digital_course/LVS

LVS

2-to-1 mux:



Schematic



Layout

```
DRC started at Wed Dec 6 21:26:15 2023

Validating hierarchy instantiation for:
library: Stanford
cell: 2-to-1mux
view: layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaDRC.rul.
Inclusion limit is set to 1000.
Running layout DRC analysis
Flat mode
Full checking.
DRC started.....Wed Dec 6 21:26:15 2023
  completed ....Wed Dec 6 21:26:15 2023
    CPU TIME = 00:00:00  TOTAL TIME = 00:00:00
***** Summary of rule violations for cell "2-to-1mux layout" *****
    Total errors found: 0
```

DRC

```

Extraction started at Wed Dec 6 21:26:50 2023

Validating hierarchy instantiation for:
library: Stanford
cell: 2-to-1mux
view: layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaEXT.rul.
Inclusion limit is set to 1000.
Running layout Extraction analysis
flat mode
Full checking.
For layer nChannelTran :
    10 shapes encountered.
    10 nmos4 ivpcell NCSU_Analog_Parts devices well formed.
For layer pChannelTran :
    10 shapes encountered.
    10 pmos4 ivpcell NCSU_Analog_Parts devices well formed.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
Extraction started.....Wed Dec 6 21:26:50 2023
    completed ....Wed Dec 6 21:26:50 2023
    CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
***** Summary of rule violations for cell "2-to-1mux layout" *****
    Total errors found: 0

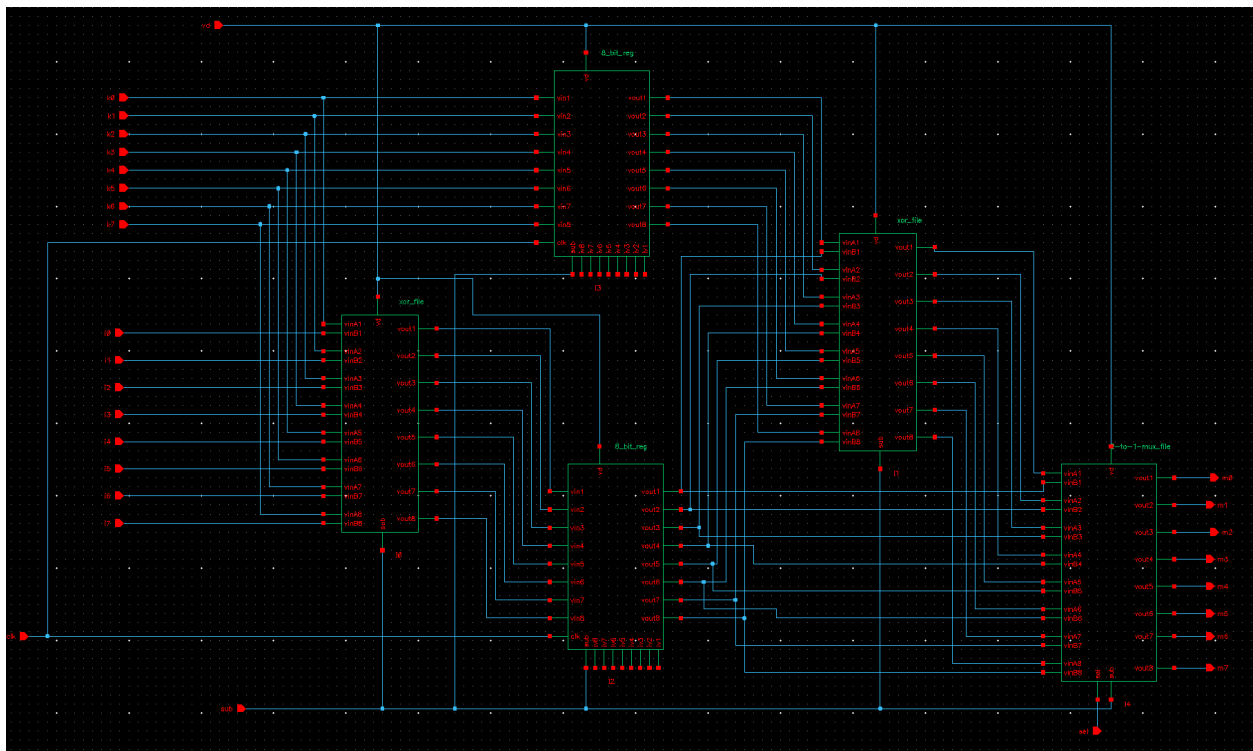
saving rep Stanford/2-to-1mux/extracted
Getting layout proptert bagGetting layout proptert bag
*Error* eval: unbound variable - ivLVSContentsDifferForm
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/eceguest/digital_course/LVS

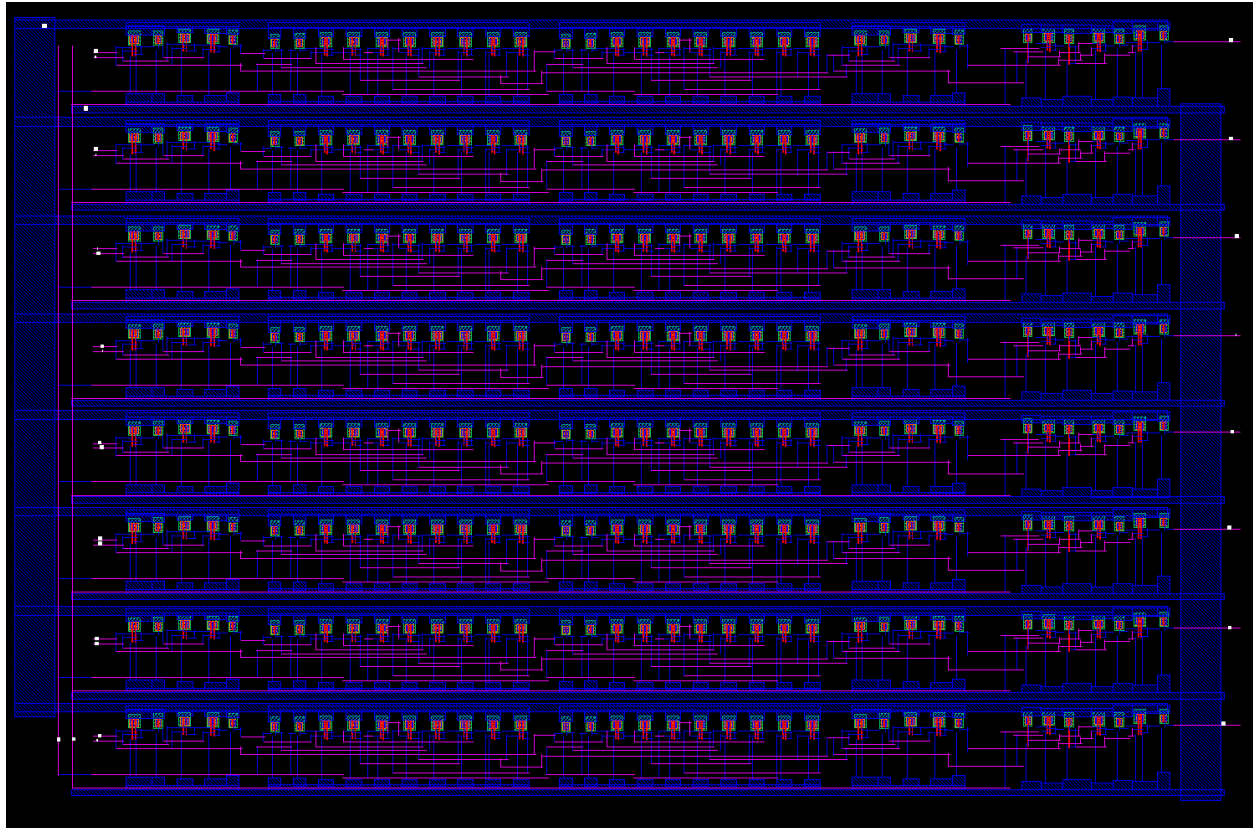
```

LVS

Encryptor:



Schematic



Layout

```
DRC started at Wed Dec 6 21:32:13 2023

Validating hierarchy instantiation for:
library: Stanford
cell: xor_encryptor_decryptor
view: layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaDRC.rul.
Inclusion limit is set to 1000.
Running layout DRC analysis
Flat mode
Full checking.
DRC started.....Wed Dec 6 21:32:13 2023
  completed ....Wed Dec 6 21:32:14 2023
  CPU TIME = 00:00:00 TOTAL TIME = 00:00:01
***** Summary of rule violations for cell "xor_encryptor_decryptor layout" *****
Total errors found: 0
```

DRC

Extraction started at Wed Dec 6 21:33:19 2023

Validating hierarchy instantiation for:

library: Stanford

cell: xor_encryptor_decryptor

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaEXT.rul.

Inclusion limit is set to 1000.

Running layout Extraction analysis

flat mode

Full checking.

For layer nChannelTran :

496 shapes encountered.

496 nmos4 ivpcell NCSU_Analog_Parts devices well formed.

For layer pChannelTran :

496 shapes encountered.

496 pmos4 ivpcell NCSU_Analog_Parts devices well formed.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

Extraction started.....Wed Dec 6 21:33:19 2023

completedWed Dec 6 21:33:20 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "xor_encryptor_decryptor layout" *****

Total errors found: 0

saving rep Stanford/xor_encryptor_decryptor/extracted

Getting layout proper bagGetting layout proper bag

LVS job is now started...

The LVS job has completed. The net-lists match.

LVS